



Rehan Khan
Systems & Control Engineering
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25M2006
M.Tech.
Gender: Male
DOB: 30/01/2001

Examination	University	Institute	Year	CPI / % Credits
Post Graduation	IIT Bombay	IIT Bombay	2026	6.21 Core: 26 Total: 38
Graduation	Parul University	Parul University	2023	7.43
Graduation Specialization: Electronics and Communication Engineering				

SCHOLASTIC ACHIEVEMENTS

- Secured **AIR-30** in UPSC NDA/NA 2018 among **450,000+** applicants; ranked **16th** in written, **62nd** in SSB
- Selected in **RIMC** through national-level exam, filling the **dual vacancy** from my state among thousands
- Earned **Code in Place 2024** certificate from **Stanford University**; invited back as **Section Leader** for CIP'25
- Achieved **Outstanding** grade across **3** semesters of **Arabic** as a Foreign Language at NDA

AWARDS & RECOGNITIONS

- Honoured twice by the **President of India** with the Chief of Staff Banner for Kilo Sqn at NDA
- Received graduation certificate from **India's first Chief of Defence Staff**, Late Gen. Bipin Rawat, in RIMC's inaugural honoured batch
- Selected to represent **NDA** at the **76th CCI Western India Squash Open 2019** (U-19 National Circuit, Mumbai), competing in the national draw from a pool of **2,000+** cadets
- Secured **1st position** at "Merge It Till You Upsurge It" — M&A case competition by LSR, Delhi University, outperforming **10+** teams including top **IIMs**
- Ranked **3rd** nationally in **CrossWeb**, a **competitive web-development** event at TechGenX 2022 (Christ University), outperforming teams across India's technical fest circuit
- Selected into **IIT Bombay IDEAS Level 1 Cohort** (Cohort 19) for a portable cooling appliance venture

KEY TECHNICAL PROJECTS

Workflow Orchestration Engine

Self Project

TypeScript, Node.js, Redis, PostgreSQL, Docker, Kubernetes

- Architected** a distributed workflow orchestration engine with **DAG-based dependency resolution**, asynchronously scheduling **100+ tasks across 20+ concurrent workflows** through Redis-backed queues.
- Designed** an asynchronous **producer-consumer architecture** with dedicated Redis step/result queues, sustaining **50+ task executions/minute** while maintaining authoritative workflow and step state.
- Designed and exposed a **RESTful API** and **CLI** for workflow submission, real-time step-level status tracking, and cancellation, powered by **Redis Pub/Sub** for live progress updates across **20+ concurrent workflows**.
- Implemented** dependency-aware **parallel execution** across **4 Kubernetes workers**, achieving **2.5x+ speedup** and **50%+ lower workflow latency** versus sequential execution.
- Engineered** fault-tolerant execution with **worker leases, heartbeats, retries and failure propagation**, recovering **90%+ of injected worker failures** within **10 seconds**.
- Integrated** the engine as the execution backbone for **AgentOS**, transforming long-running multi-step agent tasks into **fault-tolerant, dependency-resolved DAG jobs** with asynchronous distributed execution.
- Built** persistent and idempotent execution-state management with **PostgreSQL transactions**, validating duplicate-event handling and backend-restart recovery across **50+ reliability tests**.
- Benchmarked** the orchestration pipeline across **100+ DAG workloads / 500+ workflow steps**, measuring throughput, queue latency, worker utilization and recovery time to identify scheduling bottlenecks.

AgentOS - Evaluation-Driven AI Agent Runtime

Self Project

Python, LangGraph, PostgreSQL, Redis, OpenTelemetry, Langfuse, React/Next.js

- Architected** distributed agent execution with **FastAPI**, **Redis** queues and async workers; processed **200+** multi-step workflows at **98%+ task completion** under controlled load
- Engineered** fault-tolerant workflows with **Redis-backed checkpoints**, idempotency & retry/timeout policies; achieved **95%+ recovery** across injected failures
- Designed PostgreSQL + Redis** state/caching layer for task metadata, checkpoints, results & coordination; processed **500+** queued jobs with **40% fewer repeated DB/API operations**
- Built Next.js + TypeScript** execution dashboard with **REST APIs** and **WebSocket/SSE** streaming for real-time task status and worker events across concurrent runs
- Built hybrid RAG** with **BM25**, dense retrieval & reranking over **500+ PDFs**; improved retrieval recall **62%→89%** and benchmarked **4 LLMs** on success, grounding, latency & cost
- Containerized** and deployed with **Docker + CI/CD**; added automated testing, structured logging & failure-injection benchmarks, achieving **99%+ successful execution** under concurrent workloads

Large Batch vs. Small Batch Training: Generalization Tradeoffs

[Aug '25 – Nov '25]

Course Project (CS 725) – Instructor: Prof. Abir De, CSE, IIT Bombay

- Benchmarked batch sizes 1–1024 across **5 datasets and 3 neural architectures** for accuracy and efficiency.
- Quantified **10–50× lower epoch time** with large batches, requiring 1.5–2× more epochs for comparable accuracy.
- Applied **linear LR scaling** at B=512, improving accuracy 82.1%→90.2% and reducing the gap 7.9%→2.8%.

Cloud-Native Code Execution & Development Platform

Self Project

Docker, Kubernetes, Node.js, TypeScript, WebSockets, Redis, AWS S3

- **Engineered** a multi-tenant platform provisioning **20+ isolated Kubernetes workspaces** for Python, Node.js & C++ workloads with CPU, memory, disk and process quotas.
- **Built** a browser-based terminal using **xterm.js, WebSockets & node-pty**, achieving **~200ms median command-response latency** across concurrent sessions.
- **Designed** an asynchronous orchestration layer with **Node.js, Redis & Kubernetes APIs**, automating Pod, Service & Ingress provisioning with **less than 10s median startup time**.
- **Implemented** persistent workspace storage using **AWS S3 & persistent volumes**, enabling automated workspace recovery across **10+ injected failure scenarios**.

Clinical Risk Prediction & Explainability Platform

Self Project (Stanford CIP)

Python, FastAPI, PostgreSQL, React, Docker, RAG, scikit-learn, XGBoost, SHAP

- **Architected** a full-stack ML platform with **FastAPI REST APIs**, React dashboard and PostgreSQL persistence, supporting **6+ endpoints** for prediction, explanations and history
- **Engineered** configurable inference workflows with validation, error handling and model versioning, achieving **less than 150ms p95 latency** across **1,000+ API requests**
- **Integrated** SHAP explanations into the inference layer, returning feature-level attributions alongside predictions and rendering interactive patient-level explanations in React
- **Containerized** API, PostgreSQL and React services with **Docker Compose**, enabling reproducible one-command deployment across the application stack
- **Integrated** a **RAG pipeline** for retrieved clinical context, evaluating grounded responses across **100+ test queries** alongside the ML inference workflow

ML-Based BCI Wheelchair System

[Mar '22 – Mar '23]

Final Year Project – B.Tech ECE, Parul University

- Developed an **SSVEP-based BCI** pipeline using 64-channel EEG from **35 subjects** for 4 wheelchair commands.
- Implemented EEG **preprocessing** with DC-offset removal and **3–50 Hz bandpass filtering** on Oz/T10 channels.
- Extracted **FFT harmonic features** & trained **SVM** with **LOOCV + grid search** for command classification.

Spin and Quantum Control Dynamics

[May '26 – Present]

Master's Thesis – Guide: Prof. Navin Khaneja, SysCon, IIT Bombay

- Modelled spin- $\frac{1}{2}$ quantum system dynamics using **Bloch sphere representation** and **Pauli matrices**; simulated **Rabi oscillations** under magnetic field actuation with **99+% numerical fidelity** in state propagation.
- Implemented **GRAPE optimal control algorithm** for quantum state preparation; benchmarked across **3 initial-target state pairs** using gradient-based pulse optimization in **Python/NumPy**.
- Extending to **Lyapunov-based feedback control** and **stochastic noise robustness** analysis; comparing **3 control protocols** (GRAPE, CRAB, Lyapunov) on **fidelity, convergence, and noise tolerance**.

LEADERSHIP EXPERIENCE

National Defence Academy (NDA)

- Completed **Camp Rovers**, World's toughest camps in a 5-day extreme survival
- Led the training & development of **100+ junior cadets** over **6 months**, taking ownership of their conduct, training, sports & examinations
- Selected for the **Flag Corporal** shortlist — a ceremonial & mentorship appointment held by only **2 cadets** academy-wide — ranking among the top **20-30** of **300+ batchmates**
- Completed **Camp Greenhorn**, securing **3rd rank** among **18 squadrons** in a 4-day survival & endurance exercise

Rashtriya Indian Military College (RIMC)

- Appointed as **Dorm Sub NCO**, taking charge of **15+ junior cadets** across sports, discipline & day-to-day conduct

POSITIONS OF RESPONSIBILITY

PG Sports Coordinator — IIT Bombay

[Aug '25 – Aug '26]

- Coordinated **Volleyball & Badminton** tournaments end-to-end through the finals, progressing from scorer to referee & officiating both roles simultaneously in select matches
- Officiated as **umpire** for Cricket matches, ensuring fair play & accurate record-keeping across tournaments

EXTRACURRICULAR ACTIVITIES

- Selected among **40+ members** for the **IIT Bombay Hockey Team**, representing IITB at Spardha, IIT (BHU)'s official sports fest, reaching the **quarter-finals**
- Coordinated & supervised **1,000+ students** for **Engineers' Day** at Parul University
- **Second Runner-up**, FinQuest: The Finance Quiz Competition — hosted by VGSOM, IIT Kharagpur
- Selected for **Inter-IIT Sports** in Hockey